

# Lesson 8: Resonance

Frequency, exchange, and damping

Stop line and roles

**Learner boundary:** isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

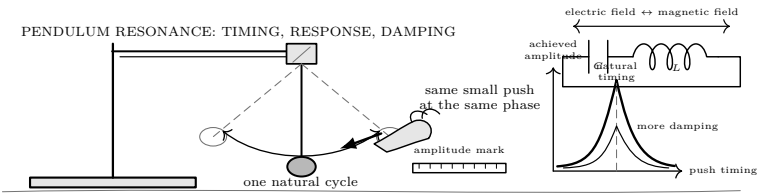
What you need

- String pendulum
- Small mass
- Ruler

- Phone stopwatch used away from apparatus
- Optional low-voltage signal source operated by adult

Predict first

Will pushes at any timing build the same amplitude?



## Investigate

1. Measure ten free oscillations and calculate period.
2. Apply equal small pushes once per cycle; count cycles needed to reach a marked amplitude.
3. Repeat with pushes deliberately early or late.
4. Add gentle damping with a paper vane and repeat.

## Explain from the evidence

Resonance is a large response when drive timing matches a system's natural exchange and losses are modest. In an ideal LC circuit, energy alternates between the capacitor's electric field and inductor's magnetic field. Resistance dissipates energy and broadens/reduces the response. Resonance selects and amplifies response; it does not create energy.

## Change one thing

Make a qualitative response curve: drive timing on the horizontal axis, achieved amplitude on the vertical.

Evidence record

Prediction

One change

Observation

Claim + because

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.